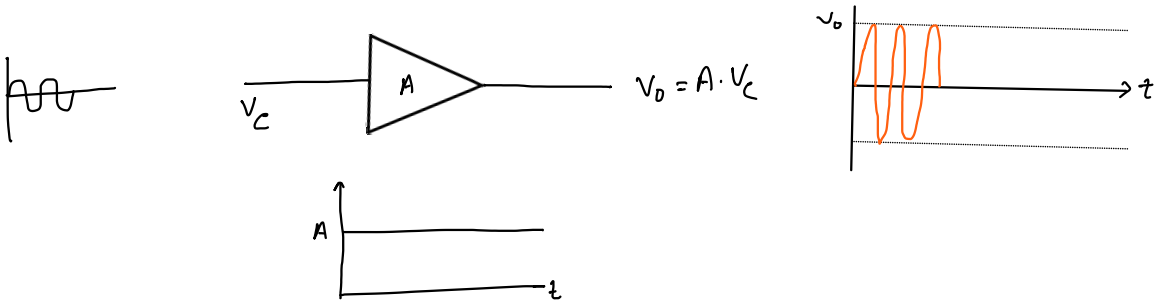
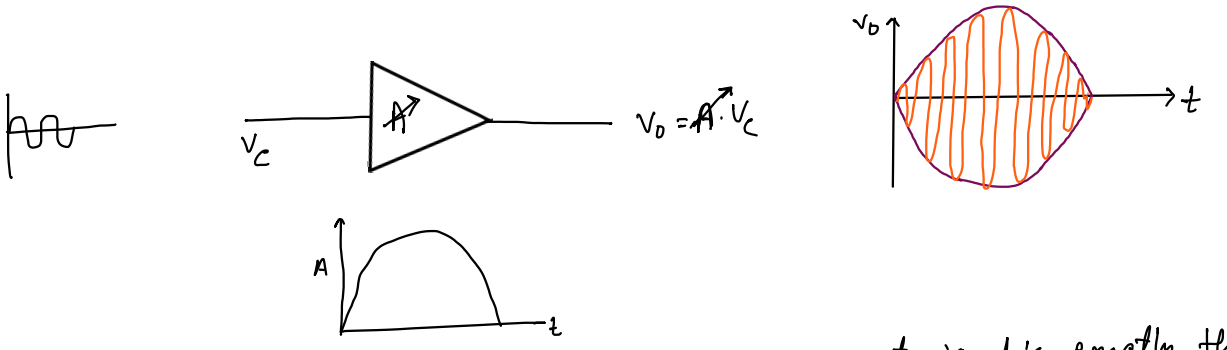
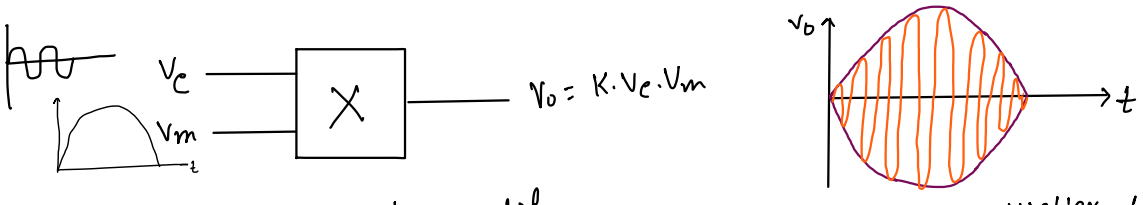
In AM, the amplitude of the carrier wave is varried in accordance with the modulating signal while the frequency of the carrier wave remains constant.



Here we use an amplifier with a constant gain A. where we feed a signal which is leveled as V_c that is carrier signal. It is a high frequency periodic signal. It runs to the amplifier which has constant amplitude. So, the outputput will be amplified and product of gain and input signal.



Now we use an amplifier with variable gain. The input signal is exactly the same periodic sinewave. But as the gain is variable, it reaches maximum and then goes to 0 then output will be varying which will be varying gain times input signal. $V_o = A \cdot V_c$
So, output voltage will be different at different points.



Instead of using varying gain amplifier, we can use analog multiplier to effectively multiply one signal (carrier signal) with another signal (modulating signal). The amplitude of carrier signal gets amplified but its frequency remains constant.

Let, $E_c = E_{cp} \cdot \sin(2\pi f_c t)$ $\longrightarrow$ carrier wave

$E_m = E_{mp} \cdot \sin(2\pi f_m t)$ $\longrightarrow$ modulating wave

$$V_o = \frac{E_m \cdot E_c}{10} = \frac{E_{cp} \cdot E_{mp}}{10} \sin(2\pi f_c t) - \sin(2\pi f_m t)$$

$$= \frac{E_{cp} \cdot E_{mp}}{20} \left[\cos 2\pi (f_c - f_m) t - \cos 2\pi (f_c + f_m) t\right]$$

$f_c - f_m$ $\rightarrow$ lower site band

$f_c + f_m$ $\rightarrow$ upper site Band

For balanced modulator $\rightarrow$ $V_{ob} = E_c \times E_m$

For standard modulator $\rightarrow$ $V_{os} = V_c(t)\left(V_c + V_m(t)\right)$